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Pediatric Tracheostomy: A Large Single-Center Experience

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Objectives: To describe the epidemiology, specifically the indications, complications, and outcomes, of pediatric tracheostomies performed in one tertiary referral unit.

Methods: Single-center retrospective cohort study of pediatric patients undergoing tracheostomy between May 2010 and May 2018 at the Newcastle upon Tyne Hospitals, United Kingdom.

Results: One hundred seventy-two pediatric tracheostomies were performed during the study period with a median age of 141 (interquartile range [IQR] 51–484) days. The most common primary indication was long-term ventilation (38.4%, 66 of 172), followed by weaning from ventilation in cardiac patients (22.1%, 38 of 172). Only 5.2% (9 of 172) of our cohort underwent tracheostomy for subglottic stenosis. The vast majority of tracheostomies were performed electively, with just 6.4% (11 of 172) performed as an emergency procedure. Early and late complication rates were 9.8% (15 of 153) and 40.0% (61 of 153), respectively. Tracheostomy decannulation was successful in 44.4% of children (68 of 153). The median duration the tracheostomy was in situ was 397 (IQR 106–708) days. All-cause mortality was 22.1% (38 of 172), with tracheostomy-related mortality at 1.2% (2 of 172).

Conclusion: We report one of the largest contemporary case series of pediatric tracheostomies. Present-day pediatric tracheostomy is primarily performed as an elective procedure in ventilated children under the age of 1 year. Pediatric tracheostomy should be considered as a long-term intervention in many children. Nevertheless, a large proportion of children are ultimately decannulated. It is important to acknowledge the significant morbidity associated with this intervention and the small-but-present risk of tracheostomy-related mortality.

Key Words: Pediatric tracheostomy, pediatric airway, cardiothoracic transplant, long-term ventilation.

Level of Evidence: 4

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INTRODUCTION

Pediatric tracheostomy can be a potentially lifesaving surgical intervention, which nevertheless is associated with significant morbidity.[1–7] Historically, the primary indication for tracheostomy in children was acute airway obstruction secondary to inflammation, such as in acute epiglottitis, diphtheria, or croup.[8,9] Pediatric vaccination programs have successfully reduced the incidence of these conditions and the need for pediatric tracheostomy in high-resource settings.[8] Improved neonatal outcomes have resulted in long-term ventilation and weaning from ventilation now being the most common indications for pediatric tracheostomy.[8,10–15]

Tracheostomy in the pediatric population can be more technically demanding than in adults due to the size of the patient.[16,17] Pediatric tracheostomy is associated with significantly more complications than adult tracheostomy and has been identified as an area requiring quality improvements.[6,16,18]

Complications result in an extended hospital stay and frequent hospital visits after discharge, increased healthcare costs, and a negative impact on quality of life for children and their carers.[1,3,19]

The procedure is often performed in children with multiple comorbidities, and overall mortality for children undergoing tracheostomy has been reported as high as 40%.[14,19–24]

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Direct tracheostomy-related mortality is reported in the range of 0.7% to 6%, dependent on patient group and follow-up duration.[14,19–24]

It is imperative that clinicians have up-to-date outcome data to appropriately counsel families about the safety and efficacy of pediatric tracheostomy and also to allow benchmarking of outcomes between institutions. Between 2000 and 2013, a large number of case series were published documenting institutional practice and comprehensive outcomes in pediatric tracheostomy.[8,10,11,14,20,24–26] In one case, this included a multicenter study.[15] This built on a number of much older series published.[9] More recently, larger scale databases studies have been published based on historic data.[27,28] A number of recent studies have also specifically investigated aspects of tracheostomy safety and compilations.[4,16,19,23,29] Few comprehensive studies of tracheostomy practice and outcomes have been published in the last 5 years.[12] Older studies are less relevant given the substantial change in indications and practice over recent years. Nationally and internationally, tracheostomy in children is a relative uncommon procedure, with only 329 procedures performed in England between 2016 and 2017.[30] In the absence of current, comprehensive, national, or international databases, the best way to establish these data sets is with large series. We aim to present a large contemporary series from a single institution that captures the safety and efficacy of pediatric tracheostomy.

MATERIALS AND METHODS

We performed a retrospective review of all pediatric tracheostomies undertaken between May 2010 and May 2018 at the Newcastle upon Tyne Hospitals National Health Service (NHS) Foundation Trust, Newcastle upon Tyne, United Kingdom. These include the Great North Children's Hospital (GNCH), a tertiary referral center for pediatrics and pediatric otolaryngology in the North East of England, and the nearby Freeman Hospital, a quaternary specialist pediatric cardiothoracic unit. The project was registered with the Clinical Governance Department of the Newcastle upon Tyne Hospitals NHS Foundation Trust.

Patients were identified from hospital clinical coding databases. Coding data is collected on every patient encounter in the NHS and characterizes multiple aspects of clinical care, including diagnoses, interventions, and outcomes. We utilized this data to identify specific procedures performed. We searched the data using the codes "tracheostomy," "temporary tracheostomy," "permanent tracheostomy," and "attention to artificial airway openings, not otherwise specified." Returned records were limited to children (≤ 16 years old) at the time of tracheostomy. The coding data was then cross-referenced with local pediatric respiratory/airway audit records of patients known to have a tracheostomy in situ. Tracheostomy was confirmed, and further data was extracted from electronic medical records for variables including age, sex, indication for tracheostomy, preoperative ventilation status, duration of tracheostomy, postoperative complications, and mortality (Table I).

Complications were divided into early postoperative and late postoperative, as previously described.[12] Early postoperative complications were defined as complications occurring within the first 48 hours after tracheostomy. Late postoperative were complications developing at any subsequent point until decannulation.[12]

Statistical analysis was performed on Microsoft Excel 14.6.5 (Redmond, WA) and Prism 7.0a for Mac (GraphPad software, CA). Quantitative variables were presented as the median and interquartile range (IQR). Nonparametric data were analyzed by Mann-Whitney U test. A P value less than 0.05 was considered statistically significant.

TABLE I.
Patient Demographics.

Demographics	Number of Patients	% of Cohort
Sex		
Male	96	55.8
Female	76	44.2
Age at tracheostomy insertion		
Below 1 year	119	69.2
Above 1 year	53	30.8
Tracheostomy insertion		
Elective	161	93.6
Emergency	11	6.4
Preoperative ventilation status		
Ventilated	128	74.4
Not ventilated	44	25.6

(n = 172)

RESULTS

Technique Description

All pediatric tracheostomies were performed under general anesthesia. A standard pediatric tracheostomy procedure was used. This includes a horizontal skin incision and vertical tracheal incision. Stay sutures are placed on either side of the tracheal incision. Stomal maturation sutures are placed from trachea to skin. The tracheostomy tube is then inserted. All postoperative patients are admitted to the pediatric intensive care unit (PICU). A routine tracheostomy tube change is performed in PICU at day 7 postoperatively. All tracheostomies were performed by a senior otolaryngologist or a training grade otolaryngologist under direct senior supervision. The majority of procedures were performed by one of four otolaryngologists with a specialist pediatric practice. A small number were also performed by a larger pool of nonpediatric otolaryngologists who cover emergency on-call operating.

Patient Demographics

A total of 182 patients were identified over the 8-year study period. Records were available for 172 patients (94.5%). Medical records pertaining to tracheostomy insertion were unavailable for four patients, and a further six patients were transferred to our care with a tracheostomy already inserted from a different institution (Fig. 1). Follow-up information was available for all children with ongoing care within Newcastle Upon Tyne Hospitals until July 2018, with a median follow-up period of 4 years 55 days (range 2 months–8 years 37 days).

Demographic information can be found in Table I. Age at time of tracheostomy ranged from day of birth to 16 years (IQR 51–484 days), 69.2% (119 of 172) patients underwent

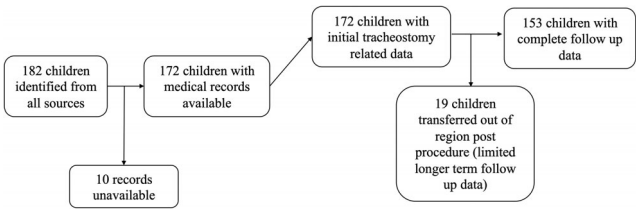

Fig. 1. Flow diagram of data availability.

tracheostomy insertion before 1 year of age (Fig. 2), with 3.5% (6 of 172) on the day of their birth, of whom two had a tracheostomy performed as part of a planned ex utero intrapartum treatment (EXIT) procedure. Indications for the two EXIT procedures were an antenatal diagnosis of an obstructive neck mass and an antenatal diagnosis of intrauterine growth retardation with multiple skeletal abnormalities, including micrognathia.

Ventilation Status and Indications for Tracheostomy

Most patients were intubated prior to tracheostomy (Table I). The median length of intubation prior to tracheostomy was 13 (IQR 7–31) days. 93.6% (161 of 172) were performed electively, whereas 6.4% (11 of 172) were performed as an emergency. Indications for tracheostomy were varied (Table II); the most common primary indication was for long-term home ventilation (66 of 172, 38.4%), followed by weaning from ventilation in cardiac patients (38 of 172, 22.1%). The primary indications in the emergency tracheostomy subgroup were firstly upper airway obstruction in early life/at birth due to syndrome craniofacial abnormalities (8 of 11). This was followed by upper airway obstruction at birth secondary to a head-and-neck mass (2 of 11) and infection in an infant with subglottic stenosis (1 of 11).

Number of Tracheostomies

The total number of tracheostomies performed within the study period was 172, an average of 22 (range 12–32) per year with the indications and number performed annually remaining constant over the 8-year period (Fig. 3).

TABLE II.
Primary Indications for Pediatric Tracheostomy.

Primary Indication	Number of Patients (%)
Long-term ventilation*	66 (38.4)
Weaning in cardiac patients	38 (22.1)
Laryngomalacia ± pharyngomalacia	13 (7.6)
Craniofacial abnormality causing upper airway obstruction	11 (6.4)
Chronic lung disease of prematurity	10 (5.8)
Subglottic stenosis	9 (5.2)
Bilateral vocal cord palsy	5 (2.9)
To facilitate mandibular surgery	5 (2.9)
Neck mass causing airway obstruction	4 (2.3)
Acute neurological insult	3 (1.7)
Obstructive sleep apnea	3 (1.7)
Chronic neurological disorder	2 (1.2)
Recurrent difficult intubation	2 (1.2)
Other	1 (0.6)

*Including tracheobronchomalacia, chronic neurodevelopmental disorder, congenital diaphragmatic hernia, congenital muscular dystrophy, congenital hypoventilation syndrome, esophageal atresia, and trachea-esophageal fistula.
(n = 172)

Length of Stay and Decannulation

The Freeman Hospital is one of two pediatric cardiothoracic transplant centers in the United Kingdom and receives a number of patients from other tertiary pediatric centers for transplant and other specialist procedures. Care of these patients is usually transferred back to the referring centers after the immediate postoperative period. Consequently, 19 patients (19 of 172, 11.0%) undergoing tracheostomy were transferred for ongoing care to other tertiary centers in the United Kingdom. These 19 children were excluded from subsequent analyses as follow-up was continued at other institutions (Fig. 1). In the remaining

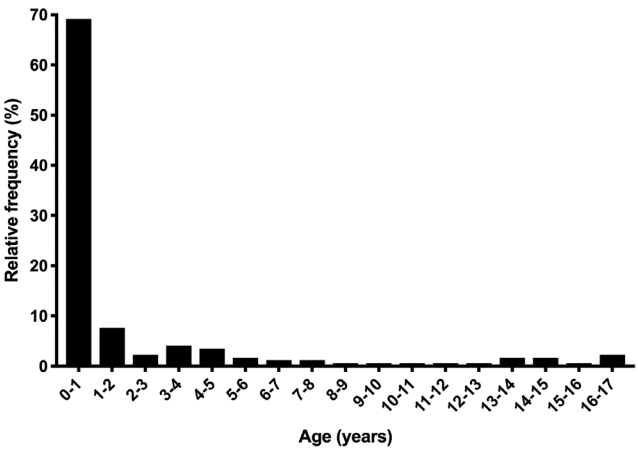

Fig. 2. Age at time of tracheostomy.

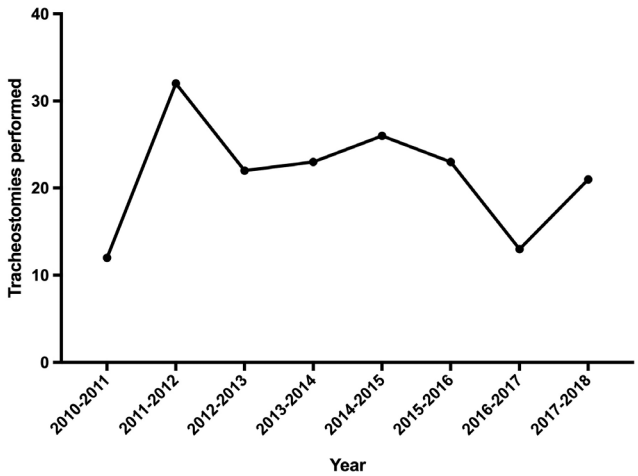

Fig. 3. Tracheostomy procedures performed over time.

patients, median length of stay of admission in which the tracheostomy was performed was 119 (IQR 41–195) days. Tracheostomy decannulation was successful in 44.4% of children (68 of 153). The median duration the tracheostomy was in situ was 397 (IQR 106–708) days.

Complications and Mortality

We identified 16 early postoperative complications occurring in 15 of 153 children (9.8%) and 72 late complications occurring in 61 of 153 children (40.0%). The most common early complication was tracheostomy tube displacement 3.9% (6 of 153), followed by bleeding 3.3% (5 of 153), respiratory distress 2.6% (4 of 153), and pneumothorax 0.7% (1 of 153). The most common late complication was the development of granulation tissue requiring intervention 19.6% (30 of 153), followed by development of a tracheocutaneous fistula at the tracheostomy site after decannulation 19.0% (29 of 153), wound infection 5.2% (8 of 153), and subglottic stenosis 2.0% (3 of 153). It should be noted that in all cases of subglottic stenosis that were found posttracheostomy, the child had already had multiple intubations prior to tracheostomy. The rate of persistent tracheocutaneous fistula after successful decannulation was 42.6% (29 of 68). The duration of cannulation was significantly ($P < 0.0001$) longer (median 693 [IQR 401–1,071] days) in children developing a fistula, compared with children who did not develop a fistula (median 227 [IQR 23–471] days). No other risk factors for tracheocutaneous fistula were identified.

Of the children who underwent tracheostomy at our institution, 22.1% (38 of 172) died. The cause of death was known for 81.2% (31 of 38) of children. Of the seven children whose cause of death was unknown, 85.7% (6 of 7) were due to care being transferred to other hospitals and therefore no longer under our follow-up, and the final case was referred to the coroner and a report is awaited. The most common causes of death were respiratory failure 4.1% (7 of 172), sepsis 3.5% (6 of 172), and complications of congenital heart disease 2.9% (5 of 172) (Table III). 1.2% (2 of 172) of the children died due to tracheostomy-related complications. Both cases were due to out-of-hospital cardiac arrests secondary to blocked tracheostomy tubes.

TABLE III.
Causes of Death.

Cause of Death	Number of Patients (% of deceased, % of cohort)
Respiratory failure	7 (18.4, 4.1)
Unknown	7 (18.4, 4.1)
Sepsis	6 (15.8, 3.5)
Congenital heart disease complications	5 (13.2, 2.9)
Cardiac complications of underlying chronic disease	3 (7.9, 1.7)
Tumor recurrence despite treatment	2 (5.3, 1.2)
Chromosomal abnormalities	2 (5.3, 1.2)
Direct complication of treatment	2 (5.3, 1.2)
Heart transplant rejection	1 (2.6, 0.6)
Multi-organ failure	1 (2.6, 0.6)
Reorientation of care due to very poor quality of life	1 (2.6, 0.6)
Smith-Lemli-Opitz Syndrome complications	1 (2.6, 0.6)
Total	37 (100, 21.6)

(n = 172)

Heart Transplant/Cardiac Disease Subgroup

Children undergoing cardiothoracic procedures and transplant often require a tracheostomy to help facilitate their weaning from ventilation.[30] The Freeman Hospital is one of two pediatric cardiothoracic transplant centers in the United Kingdom, and therefore a large proportion (22.1%, 38 of 172) of the tracheostomies were performed on these children. Because this was a large group with specific comorbidities, we performed a subgroup analysis on this population. Many of these children are temporarily transferred to our institution for transplant, and therefore follow-up information was only available for 29 of 38 (74.4%) of these children. They had a tracheostomy for a median of 629 (IQR 134–1,493) days. In these 29 cases, all children were intubated prior to elective tracheostomy for a median of 23 (IQR 11–32) days. The median age at tracheostomy in the children undergoing cardiothoracic procedures was 211 (IQR 123–560) days. This is older than the median age of 111 (IQR 47–343) days in children requiring tracheostomy for other indications. The median length of stay in children with cardiothoracic diagnoses was 156 (IQR 67–275) days, compared with a shorter 109 (IQR 33–188) days in the remainder of the cohort. The median time to successful decannulation in the cardiothoracic subgroup was 143 (IQR 77–347) days, with 51.7% (15 of 29) successfully decannulated. Complication rates were comparable with the general cohort, albeit that mortality was higher in the cardiac group at 31% (9 of 29) versus 22.8% in the noncardiac group (29 of 127). Due to the small subgroup population, statistical analysis was not performed.

DISCUSSION

We report one of the largest contemporary cohorts of pediatric tracheostomies, with 172 tracheostomies performed over an 8-year period. In keeping with other published studies, the majority of patients (69.2%) had their tracheostomy performed before 1 year of age.[8,10,12,15,31] Tracheostomy was performed in intubated patients in 74.4% of cases, who were ventilated a median 13 days prior to procedure. The most common primary indication was long-term ventilation (38.4%), followed by weaning from ventilation in cardiac patients (22.1%). Previous studies have reported that long-term ventilation was the most common indication for tracheostomy (range 20%–61%), followed by subglottic stenosis (range 14%–36.8%).[10,12,32–35] The high rates of tracheostomy for weaning from ventilation in cardiac patients are likely related to our unit's status as a quaternary referral center for pediatric cardiac surgery. However, it is interesting that only 5.2% of our cohort underwent tracheostomy for subglottic stenosis as the

primary indication, which may be a consequence of improvements in neonatal airway management and the greater application of modern endoscopic techniques to treat airway stenosis in recent years.[36–38] Tracheostomies were primarily planned procedures, with only 6.4% performed as an emergency; this is in keeping with previous published emergency tracheostomy rates (range 6%–12%) but again highlights the recent changes in indications of pediatric tracheostomy from airway obstruction to long-term ventilation.

In our series, 44.4% of children were successfully decannulated; this is higher than other reported series (32%–39%). This may be attributed to the longer follow-up in this study (a median of 4 years 55 days).[8,10,12,31] Within our series, the tracheostomy remained in situ a median 397 (IQR 106–708) days, which is comparable to the average durations of tracheostomy in the literature (12–28 months).[8,10,12]

We report an early complication rate of 9.8% and a late complication rate of 40.0%. Comparable series report between 7.4% to 11.2% early complications and 10% to 51% late complications.[10,14,16,19,20,23] Our most common early complication was tube displacement (4%), and the most common late complication was granulation tissue requiring intervention (19.6%), similar to other series.[8,25,39] Whereas these rates are comparable to other studies, they highlight the significant burden of complications that these children and their families face after a tracheostomy, including in some cases further operations. It has previously been highlighted that complications result in increased healthcare costs and a negative impact on quality of life for children and their carers.[1,3,19] Reduction in the rates of these complications should be a key target for research and quality improvement programs.

In our series, 22.1% of children died; this is similar to that reported in the literature (15%–39.2%).[8,10,12,35,40,41] The most common causes of death in our cohort were respiratory failure, sepsis, or complications of underlying congenital heart disease. Two of these deaths (1.2%) were related to tube blockage and subsequent fatal complications and therefore deemed a tracheostomy-related death (literature reported range 0.7%–6%).[14,19–24] It should be noted that these deaths occurred out of hospital. There were no tracheostomy-related deaths within our institution during the study period. These fatalities and neurological complications secondary to tube blockage identified in our series are potentially avoidable and identify the need for better support in the community for carers of children who have tracheostomies.

Approximately one-fifth of our cohort were children with cardiothoracic comorbidities such as complex congenital heart disease requiring multiple cardiac surgeries, with some inevitably requiring transplantation who underwent a tracheostomy to enable weaning from ventilation. This is one of the largest series in this patient group and uniquely also includes a corresponding noncardiac cohort from the same institution for comparison. This cardiac cohort differs in many ways to the noncardiac tracheostomized children. All tracheostomies in this cardiac subgroup were performed electively in exclusively intubated children. The median duration of intubation prior to tracheostomy was 23 days, which is similar to the 30-day duration described by Hoskote et al. series (London, U.K.) in a series of 37 children with underlying congenital heart disease.[25] Interestingly, the median duration of tracheostomy in our series was 629 (IQR 134–1,493) days, which is shorter than previously reported by Hoskote et al. with a median 1,497 days (range 792–1,735 days). Hoskote et al. reported a 40.5% decannulation rate within an 18-month follow-up period.[25] Within the cardiothoracic subgroup in our study, 51.7% were successfully decannulated within a median 143 days (11.9 months) of tracheostomy. The cardiothoracic subgroup experienced minimal complications, but the mortality in this subgroup was 31% (9 of 29). Although higher than for children undergoing tracheostomy for other indications, this is similar to that reported in cardiothoracic children by Ortmann et al. (Kansas, U.S.), 28%; and Hoskote et al. series (London, U.K.), 40.5%.[25,39]

The limitations of this study included the 5.5% (10 of 182) of tracheostomy patient's medical notes that were not available and the study duration of 8 years. Other series have tracked trends in pediatric tracheostomy over longer time periods; however, our series reflects contemporary practice with more procedures performed in a shorter period. As a specialist referral center, 19 children (19 of 172, 11.0%) who were transferred temporarily to the Newcastle Hospitals were then transferred back to their original hospital, which reduced the available follow-up data (153 total children). The pediatric tracheostomy population is highly heterogeneous with multiple comorbidities. When pooling our results for outcomes, such as length of hospital stays or decannulation, this may be affected by these factors. Therefore, interpretation of these and other outcomes should consider these limitations. We aimed to where possible standardize data collection to allow pooling of data and comparison between centers within the United Kingdom and internationally.[12]

Tracheostomy, although relatively common in our institution compared to other units, is a rare procedure nationally and internationally. The procedure is associated with significant morbidity. We present a large case series from our institution that will add to the contemporary literature on this topic and allow pooling of national and international data to allow clinicians to appropriately counsel families about the safety and efficacy of pediatric tracheostomy, as well as to allow benchmarking of outcomes between institutions. We believe that the development of a national/international registry for pediatric tracheostomy with associated outcomes will enable further research in this area.

CONCLUSION

We report one of the largest recent series of pediatric tracheostomies, with 172 tracheostomies performed over an 8-year period at the Newcastle Upon Tyne Hospitals (GNCH and Freeman Hospital), United Kingdom. Present-day pediatric tracheostomy is primarily performed as an elective procedure in ventilated children under the age of 1 year. Pediatric tracheostomy should be considered as a long-term intervention in many children. Nevertheless, a large proportion of children are ultimately decannulated. Children with tracheostomies have a high burden of

complications. All-cause mortality was high, whereas specific tracheostomy-related mortality was low. Nevertheless, these deaths still highlight the need for improved community support for tracheotomized children and their families. We would advocate the need for a comprehensive national/international registry of pediatric tracheostomy outcomes in order to help compare institutional practice, identify further areas for practice improvement, and improve the current evidence base.

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